Progress Report

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Do not really know how to fit this into 2 pages.

Repository

The repository is available at https://github.com/jake-bounds/mini-internet.

Architecture

Configuration

Topology is defined in toml files inside the topology directory. They specify hosts and their router address, and routers and their next hop tables. They specify UDP ports on the host machine for both. Network conditions are defined in the conditions directory. They specify link delay and loss on a source $\rightarrow$ destination address basis. Then a config python file reads the config, throws validation errors, and makes the config available to host and router programs.

Packets

Packets store what is defined in the assignment pdf; namely, source address, destination address, message type (probe, echo), message id, payload. During transmission, packets are converted from a dataclass to bytes by storing the payload as a base64 string, then JSON encoding the dataclass, then UTF-8 encoding the JSON string as bytes. They are converted back into dataclasses by the reverse process.

Hosts

Hosts use asyncio.create_datagram_endpoint to allow for async processing of both user input to send probes to destination hosts and to accept packets from the router. Sent out probes the host is waiting on is stored in a dictionary keyed by the msg_id, along with an asyncio timer for timeout.

Router

Routers do not use asyncio, though later I will probably make them do so. For now they just keep a socket in the main thread, receive incoming packets and forward them to the next hop based on the configuration. They perform validation to determine whether to drop packets. Link delay is achieved with threading.Timer.

Defects

No defects to my knowledge. One thing I would like to do is create an actual link class for separation of concerns purposes. Maybe I could do struct packing instead of JSON for packets also.

Schedule, Extension, Experimental Plan

I will do reliable transfer. Probably implementation for the next 2 weeks, then acceptance criteria and required evidence the next week, then presentation stuff the next week.

Acceptance Tests

In the repo, you would first need to create a virtual environment. I used Python 3.14. You also need bash and to be on Linux as one test requires sending raw bytes to a UNIX udp socket file.

The core acceptance scenarios can all be performed by bash scripts inside of the acceptance_scenarios directory. These work by running the router and host processes, then feeding input into either Host 1 or Host 2 or both and then either counting logs or printing them.

Multihop

```
jake:~/mini-internet$ bash acceptance_scenarios/multihop.bash

Run Id: nXgtHD
Topology: topology/upper.toml  Conditions: conditions/none.toml  Seed: 1
Logs: logs/multihop

Host 1: 20/20 matching responses, 0 bad payloads, 0 timeouts, 20 probes echoed
Host 2: 20/20 matching responses, 0 bad payloads, 0 timeouts, 20 probes echoed
```

Configurable Paths

Upper Path

```
jake:~/mini-internet$ bash acceptance_scenarios/configurable_paths.bash

Run Id: UK5DlH
Topology: topology/upper.toml Conditions: conditions/none.toml  Seed: 1
Logs: logs/configurable_paths

== Router A
Using seed 1:Router A
UK5DlH | Router A | 1792634383.137582 | 1cf2 | probe | Host 1 | Host 2 | Router B | 5 | rece
UK5DlH | Router A | 1792634383.1376145 | 1cf2 | probe | Host 1 | Host 2 | Router B | 4 | for
UK5DlH | Router A | 1792634383.1379075 | 8904 | probe | Host 2 | Host 1 | Host 1 | 3 | recei
UK5DlH | Router A | 1792634383.1379192 | 8904 | probe | Host 2 | Host 1 | Host 1 | 2 | forwa
UK5DlH | Router A | 1792634383.1380932 | 8904 | echo | Host 1 | Host 2 | Router B | 5 | rece
UK5DlH | Router A | 1792634383.1381013 | 8904 | echo | Host 1 | Host 2 | Router B | 4 | forwa
UK5DlH | Router A | 1792634383.1381273 | 1cf2 | echo | Host 2 | Host 1 | Host 1 | 3 | receive
UK5DlH | Router A | 1792634383.1381316 | 1cf2 | echo | Host 2 | Host 1 | Host 1 | 2 | forwarc

== Router B
Using seed 1:Router B
UK5DlH | Router B | 1792634383.137748 | 1cf2 | probe | Host 1 | Host 2 | Router D | 4 | rece
UK5DlH | Router B | 1792634383.1377804 | 1cf2 | probe | Host 1 | Host 2 | Router D | 3 | for
UK5DlH | Router B | 1792634383.1378334 | 8904 | probe | Host 2 | Host 1 | Router A | 4 | rec
UK5DlH | Router B | 1792634383.1378539 | 8904 | probe | Host 2 | Host 1 | Router A | 3 | for
UK5DlH | Router B | 1792634383.1380901 | 1cf2 | echo | Host 2 | Host 1 | Router A | 4 | rece
```

```
UK5DlH | Router B | 1792634383.138098 | 1cf2 | echo | Host 2 | Host 1 | Router A | 3 | forwa
UK5DlH | Router B | 1792634383.1384206 | 8904 | echo | Host 1 | Host 2 | Router D | 4 | rece
UK5DlH | Router B | 1792634383.1384351 | 8904 | echo | Host 1 | Host 2 | Router D | 3 | forwa

== Router C
Using seed 1:Router C

== Router D
Using seed 1:Router D
UK5DlH | Router D | 1792634383.1375713 | 8904 | probe | Host 2 | Host 1 | Router B | 5 | rece
UK5DlH | Router D | 1792634383.1376143 | 8904 | probe | Host 2 | Host 1 | Router B | 4 | forw
UK5DlH | Router D | 1792634383.1378536 | 1cf2 | probe | Host 1 | Host 2 | Host 2 | 3 | receiv
UK5DlH | Router D | 1792634383.1378636 | 1cf2 | probe | Host 1 | Host 2 | Host 2 | 2 | forwa
UK5DlH | Router D | 1792634383.1380403 | 1cf2 | echo | Host 2 | Host 1 | Router B | 5 | rece
UK5DlH | Router D | 1792634383.138045 | 1cf2 | echo | Host 2 | Host 1 | Router B | 4 | forwa
UK5DlH | Router D | 1792634383.1384974 | 8904 | echo | Host 1 | Host 2 | Host 2 | 3 | receive
UK5DlH | Router D | 1792634383.138506 | 8904 | echo | Host 1 | Host 2 | Host 2 | 2 | forward
```

Lower Path

```
jake@cerberus:~/mini-internet$ bash acceptance_scenarios/configurable_paths.bash lower

Run Id: TREZrs
Topology: topology/lower.toml Conditions: conditions/none.toml  Seed: 1
Logs: logs/configurable_paths

== Router A
Using seed 1:Router A
TREZrs | Router A | 1792634446.8607237 | a7c9 | probe | Host 1 | Host 2 | Router C | 5 | rece
TREZrs | Router A | 1792634446.8607593 | a7c9 | probe | Host 1 | Host 2 | Router C | 4 | forw
TREZrs | Router A | 1792634446.8615968 | a7c9 | echo | Host 2 | Host 1 | Host 1 | 3 | receive
TREZrs | Router A | 1792634446.8616233 | a7c9 | echo | Host 2 | Host 1 | Host 1 | 2 | forward
TREZrs | Router A | 1792634446.8626266 | 5bcc | probe | Host 2 | Host 1 | Host 1 | 3 | receiv
TREZrs | Router A | 1792634446.8626366 | 5bcc | probe | Host 2 | Host 1 | Host 1 | 2 | forwa
TREZrs | Router A | 1792634446.8627634 | 5bcc | echo | Host 1 | Host 2 | Router C | 5 | rece
TREZrs | Router A | 1792634446.8627772 | 5bcc | echo | Host 1 | Host 2 | Router C | 4 | forwa

== Router B
Using seed 1:Router B
```

```
== Router C
Using seed 1:Router C
TREZrs | Router C | 1792634446.8608978 | a7c9 | probe | Host 1 | Host 2 | Router D | 4 | rec
TREZrs | Router C | 1792634446.8609219 | a7c9 | probe | Host 1 | Host 2 | Router D | 3 | for
TREZrs | Router C | 1792634446.8614905 | a7c9 | echo | Host 2 | Host 1 | Router A | 4 | rece
TREZrs | Router C | 1792634446.8615203 | a7c9 | echo | Host 2 | Host 1 | Router A | 3 | forw
TREZrs | Router C | 1792634446.8625684 | 5bcc | probe | Host 2 | Host 1 | Router A | 4 | rec
TREZrs | Router C | 1792634446.8625786 | 5bcc | probe | Host 2 | Host 1 | Router A | 3 | for
TREZrs | Router C | 1792634446.862889 | 5bcc | echo | Host 1 | Host 2 | Router D | 4 | recei
TREZrs | Router C | 1792634446.8629048 | 5bcc | echo | Host 1 | Host 2 | Router D | 3 | forw

== Router D
Using seed 1:Router D
TREZrs | Router D | 1792634446.8610532 | a7c9 | probe | Host 1 | Host 2 | Host 2 | 3 | recei
TREZrs | Router D | 1792634446.8610854 | a7c9 | probe | Host 1 | Host 2 | Host 2 | 2 | forwa
TREZrs | Router D | 1792634446.8613598 | a7c9 | echo | Host 2 | Host 1 | Router C | 5 | rece
TREZrs | Router D | 1792634446.8613694 | a7c9 | echo | Host 2 | Host 1 | Router C | 4 | forw
TREZrs | Router D | 1792634446.8624923 | 5bcc | probe | Host 2 | Host 1 | Router C | 5 | rec
TREZrs | Router D | 1792634446.862509 | 5bcc | probe | Host 2 | Host 1 | Router C | 4 | forw
TREZrs | Router D | 1792634446.8629537 | 5bcc | echo | Host 1 | Host 2 | Host 2 | 3 | receiv
TREZrs | Router D | 1792634446.8629627 | 5bcc | echo | Host 1 | Host 2 | Host 2 | 2 | forwar
```

We can see that C has no logs in upper, and B has no logs in lower.

Hop Limit

```
jake@cerberus:~/mini-internet$ bash acceptance_scenarios/hop_limit.bash

Run Id: zOeasw
Topology: topology/loop.toml Conditions: conditions/none.toml  Seed: 1
Logs: logs/hop_limit

== Router A
Using seed 1:Router A
zOeasw | Router A | 1792634537.5530741 | 65f8 | probe | Host 1 | Host 2 | Router C | 5 | rec
zOeasw | Router A | 1792634537.553109 | 65f8 | probe | Host 1 | Host 2 | Router C | 4 | forw
zOeasw | Router A | 1792634537.553397 | 65f8 | probe | Host 1 | Host 2 | Router C | 3 | rece
zOeasw | Router A | 1792634537.5534158 | 65f8 | probe | Host 1 | Host 2 | Router C | 2 | for
zOeasw | Router A | 1792634537.5535743 | 65f8 | probe | Host 1 | Host 2 | Router C | 1 | rec
zOeasw | Router A | 1792634537.5535939 | 65f8 | probe | Host 1 | Host 2 | Router C | 0 | dro
```

```
== Router C
Using seed 1:Router C
zOeasw | Router C | 1792634537.5532734 | 65f8 | probe | Host 1 | Host 2 | Router A | 4 | rece
zOeasw | Router C | 1792634537.5533 | 65f8 | probe | Host 1 | Host 2 | Router A | 3 | forwarc
zOeasw | Router C | 1792634537.5534902 | 65f8 | probe | Host 1 | Host 2 | Router A | 2 | rece
zOeasw | Router C | 1792634537.553504 | 65f8 | probe | Host 1 | Host 2 | Router A | 1 | forwa
```

Error Handling

First we send a packet of garbage bytes (not done inside of the host program as we must echo write directly to a socket file), then a packet with a nonexistent destination address, then a packet that is too large.

```
jake@cerberus:~/mini-internet$ bash acceptance_scenarios/error_handling.bash

Run Id: d3R4ev
Topology: topology/loop.toml Conditions: conditions/none.toml  Seed: 1
Logs: logs/error_handling

== Host 1
Using seed 1:Host 1
To send a probe, type <dest_addr>:<msg> and press enter.
Sending probe f444 to Host 3.
Sending probe c037 to Host 2.
No response received for probe f444 in 5 seconds.
No response received for probe c037 in 5 seconds.

== Router A
Using seed 1:Router A
d3R4ev | Router A | 1792634703.5558481 | receive,drop (failed to decode)
d3R4ev | Router A | 1792634703.6025295 | f444 | probe | Host 1 | Host 3 | null | 5 | drop (no
d3R4ev | Router A | 1792634703.6027217 | receive,drop (too large)
```

Loss and Delay

```
jake@cerberus:~/mini-internet$ bash acceptance_scenarios/loss_and_delay.bash

Run Id: QoZF2V  Seed: 1
```

```
=== forced_drop  Topology: topology/upper.toml  Conditions: conditions/forced_loss.toml
-- Router A
QoZF2V | Router A | 1792634745.907182 | 1120 | probe | Host 1 | Host 2 | Router B | 5 | rece
QoZF2V | Router A | 1792634745.907229 | 1120 | probe | Host 1 | Host 2 | Router B | 4 | drop
-- Host 1
Sending probe 1120 to Host 2.
No response received for probe 1120 in 5 seconds.

=== random_loss  Topology: topology/upper.toml  Conditions: conditions/loss.toml
Host 1: 12/20 responses, 8 timeouts. Router A: 8 drops

=== delay  Topology: topology/upper.toml  Conditions: conditions/delay.toml
-- Host 1
Sending probe 7873 to Host 2.
Sending probe 6882 to Host 2.
Sending probe 82c7 to Host 2.
Response received for probe 7873 in 0.5015176900051301 seconds.
Response received for probe 6882 in 0.5012876079999842 seconds.
Response received for probe 82c7 in 0.5011903090053238 seconds.

=== late_response  Topology: topology/lower.toml  Conditions: conditions/delay.toml
-- Router C
QoZF2V | Router C | 1792634775.9569743 | eb13 | probe | Host 1 | Host 2 | Router D | 4 | rec
QoZF2V | Router C | 1792634775.9570127 | eb13 | probe | Host 1 | Host 2 | Router D | 3 | for
QoZF2V | Router C | 1792634775.9575913 | eb13 | echo | Host 2 | Host 1 | Router A | 4 | rece
QoZF2V | Router C | 1792634775.9576132 | eb13 | echo | Host 2 | Host 1 | Router A | 3 | forw
-- Host 1
Sending probe eb13 to Host 2.
No response received for probe eb13 in 5 seconds.
Received echo for a probe that wasn't sent by us (eb13).
```

Baseline Measurements

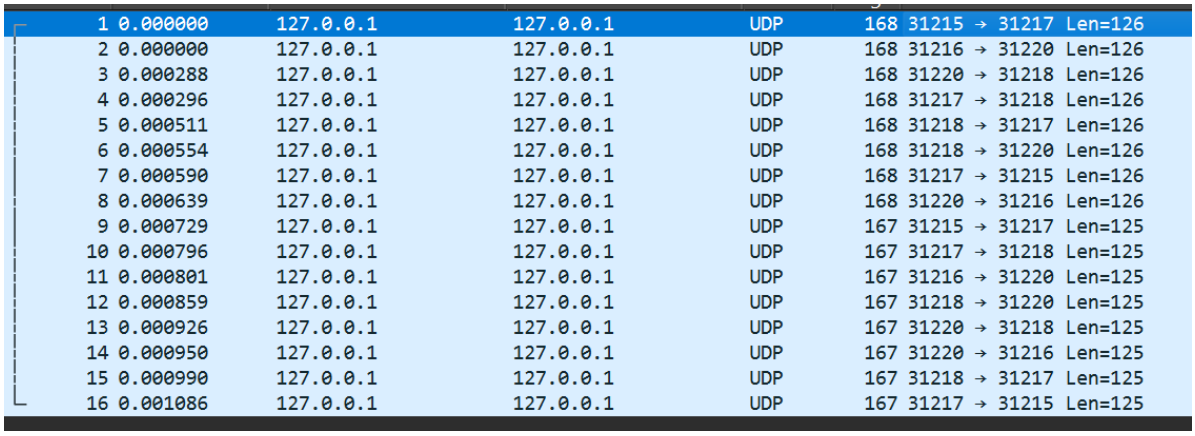
```
jake@cerberus:~/mini-internet$ bash measurements/baseline.bash

Run Id: nZ3u4Q
Topology: topology/upper.toml  Probes: 20  Interval: 0.2s  Timeout: 5s
delay0_loss0 seed 1
delay0_loss0 seed 2
```

```
delay0_loss0 seed 3
delay0_loss05 seed 1
delay0_loss05 seed 2
delay0_loss05 seed 3
delay100_loss0 seed 1
delay100_loss0 seed 2
delay100_loss0 seed 3
delay100_loss05 seed 1
delay100_loss05 seed 2
delay100_loss05 seed 3

condition                  success %          mean RTT ms        timeouts
delay0_loss0             100.0 ± 0.0           1.10 ± 0.00       0.0 ± 0.0
delay0_loss05             90.0 ± 8.7           1.13 ± 0.06       2.0 ± 1.7
delay100_loss0           100.0 ± 0.0         202.07 ± 0.06       0.0 ± 0.0
delay100_loss05           90.0 ± 8.7         202.03 ± 0.06       2.0 ± 1.7
```

Wireshark



1	0.000000	127.0.0.1	127.0.0.1	UDP	168 31215 → 31217 Len=126
2	0.000000	127.0.0.1	127.0.0.1	UDP	168 31216 → 31220 Len=126
3	0.000288	127.0.0.1	127.0.0.1	UDP	168 31220 → 31218 Len=126
4	0.000296	127.0.0.1	127.0.0.1	UDP	168 31217 → 31218 Len=126
5	0.000511	127.0.0.1	127.0.0.1	UDP	168 31218 → 31217 Len=126
6	0.000554	127.0.0.1	127.0.0.1	UDP	168 31218 → 31220 Len=126
7	0.000590	127.0.0.1	127.0.0.1	UDP	168 31217 → 31215 Len=126
8	0.000639	127.0.0.1	127.0.0.1	UDP	168 31220 → 31216 Len=126
9	0.000729	127.0.0.1	127.0.0.1	UDP	167 31215 → 31217 Len=125
10	0.000796	127.0.0.1	127.0.0.1	UDP	167 31217 → 31218 Len=125
11	0.000801	127.0.0.1	127.0.0.1	UDP	167 31216 → 31220 Len=125
12	0.000859	127.0.0.1	127.0.0.1	UDP	167 31218 → 31220 Len=125
13	0.000926	127.0.0.1	127.0.0.1	UDP	167 31220 → 31218 Len=125
14	0.000950	127.0.0.1	127.0.0.1	UDP	167 31220 → 31216 Len=125
15	0.000990	127.0.0.1	127.0.0.1	UDP	167 31218 → 31217 Len=125
16	0.001086	127.0.0.1	127.0.0.1	UDP	167 31217 → 31215 Len=125

Packets 1, 4, 6, 8 show Host 1's probe to Host 2.

Packets 11, 13, 15, 16 show the echo being sent back to Host 1.

They correspond to these logs:

```
Sending probe 2dc4 to Host 2.
pnoZns | Router A | 1792635869.3438892 | 2dc4 | probe | Host 1 | Host 2 | Router B | 5 | rec
pnoZns | Router A | 1792635869.3439243 | 2dc4 | probe | Host 1 | Host 2 | Router B | 4 | for
pnoZns | Router B | 1792635869.3441973 | 2dc4 | probe | Host 1 | Host 2 | Router D | 4 | rec
```

```
pnoZns | Router B | 1792635869.3442056 | 2dc4 | probe | Host 1 | Host 2 | Router D | 3 | forw
pnoZns | Router D | 1792635869.3442655 | 2dc4 | probe | Host 1 | Host 2 | Host 2 | 3 | receiv
pnoZns | Router D | 1792635869.3442836 | 2dc4 | probe | Host 1 | Host 2 | Host 2 | 2 | forwar
p
Received probe 2dc4. Echoing.
pnoZns | Router D | 1792635869.3445647 | 2dc4 | echo | Host 2 | Host 1 | Router B | 5 | recei
pnoZns | Router D | 1792635869.3445747 | 2dc4 | echo | Host 2 | Host 1 | Router B | 4 | forwa
pnoZns | Router B | 1792635869.3446293 | 2dc4 | echo | Host 2 | Host 1 | Router A | 4 | recei
pnoZns | Router B | 1792635869.3446405 | 2dc4 | echo | Host 2 | Host 1 | Router A | 3 | forwa
pnoZns | Router A | 1792635869.3447278 | 2dc4 | echo | Host 2 | Host 1 | Host 1 | 3 | receive
pnoZns | Router A | 1792635869.3447375 | 2dc4 | echo | Host 2 | Host 1 | Host 1 | 2 | forward
Response received for probe 2dc4 in 0.0012912569945910946 seconds.
```

Since my virtual protocol is JSON, it is easy to inspect the packets.

```
00  00 00 00 00 00 00 00 00  00 00 00 00 08 00 45 00   ........ ......E·
10  00 99 4a 60 40 00 40 11  f1 f1 7f 00 00 01 7f 00   ··J`@·@· ........
20  00 01 79 f1 79 ef 00 85  fe 98 7b 22 73 6f 75 72   ··y·y··· ··{"sour
30  63 65 5f 61 64 64 72 22  3a 20 22 48 6f 73 74 20   ce_addr" : "Host
40  32 22 2c 20 22 64 65 73  74 5f 61 64 64 72 22 3a   2", "des t_addr":
50  20 22 48 6f 73 74 20 31  22 2c 20 22 6d 73 67 5f    "Host 1 ", "msg_
60  69 64 22 3a 20 22 32 64  63 34 22 2c 20 22 68 6f   id": "2d c4", "ho
70  70 5f 6c 69 6d 69 74 22  3a 20 32 2c 20 22 6d 73   p_limit" : 2, "ms
80  67 5f 74 79 70 65 22 3a  20 22 65 63 68 6f 22 2c   g_type":  "echo",
90  20 22 70 61 79 6c 6f 61  64 22 3a 20 22 62 58 4e    "payloa d": "bXN
a0  6e 4c 57 67 78 22 7d                               nLWgx"}
```

This is port 31217 to 31215, or Router A to Host 1. Its source address is Host 2, and destination is Host 1. This is the echo sent from Host 2 to Host 1 in response to Host 1's probe.

Contributions

I am working alone